Equal Opportunity in Ranking

Shubham Singh
shusinghcs@gmail.com

Admissions to Race Academy

9 8 3 2

What is fairness?

Define a notion of unfairness or (violation of) equal opportunity.

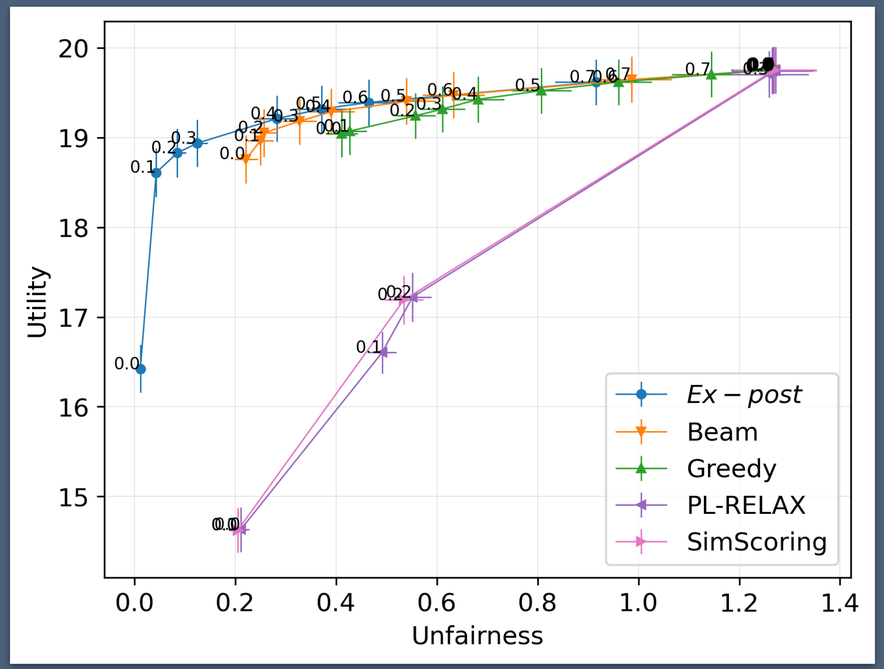
Unfairness

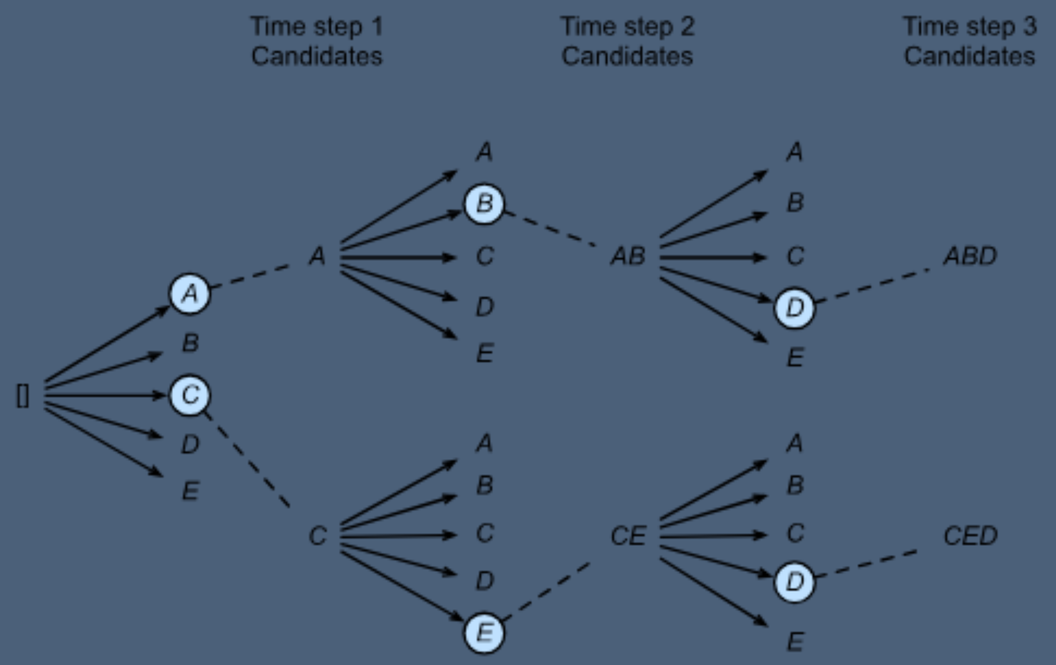
$$\left| \frac{1}{|G_1|} \sum \text{Quality} * \text{Position-Bias} - \frac{1}{|G_2|} \sum \text{Quality} * \text{Position-Bias} \right|$$

A **trade-off** strikes a balance between most efficient and least unfair ranking.

Results

- The most common way in ranking is to learn a score per document and then sort.
- This **scoring fails** to achieve an optimal trade-off!





Beam Search example: https://d2l.ai/chapter_recurrent-modern/beam-search.html